

Figure 2: Controller class integration in Scene Builder

- with your IDE, so you can close this instance for now.
- You must be wondering where Scene Builder has been installed, because during the installation process you were not asked for the installation directory. Actually, it is installed by default in the directory path `C:\Users\<your computer name>\AppData\Local\SceneBuilder`, but you can move its installation folder to any other location, depending on your convenience.
 - Now that this is done, let's shift focus to the Eclipse IDE. Start Eclipse and create any JavaFX project in it. For example, I am naming the project I'm creating 'MyFirstSceneBuilder' for the purpose of this tutorial. If you are not aware of how to create a JavaFX project in Eclipse, then you can read my previous article on JavaFX. With the JavaFX project, the application package gets automatically created with *Main.java* class and the *application.css* file; so create one more package named *application.view* in the *src* folder, where we will create a file to provide a view to our application.
 - After the package has been created in your workspace, right-click on the *application.view* package and add a 'New FXML document' to your project. If it's not available in the list, then you can find it under the *Others > JavaFX tree* section.
 - Provide a logical name to it, e.g., *Main*; then select *Border Pane* as the root element and click on 'Finish', which will add this FXML document to your project. You can view your FXML file, which will be just like an empty XML file with a `<BorderPane>` tag in it.
 - This FXML document that we have created can be opened with Scene Builder, and then we can start dragging and dropping to build our interfaces. But before that, we need to configure it with Eclipse.
 - For this, navigate to 'Windows menu > Preferences'. In the *Preferences* dialogue box, search for *JavaFX* and select it. Now you will be able to see the *SceneBuilder* executable text box on the right-hand side.
 - Browse to the location where you have copied your installed folder in Step 2. In that folder, there must be a *SceneBuilder.exe* file; select it, and click OK. Click on *Apply* in the *Preferences* dialogue box to close it.

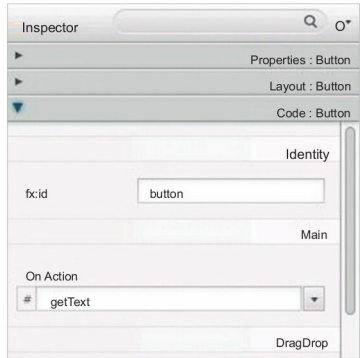


Figure 3: Setting *fx:id* and *On Action* field for 'button'

- Now go to your *Main.fxml* file in the package explorer and right-click on it. You will be able to see the option 'Open with Scene Builder' in the *Context* menu. Select it and this file will be opened inside your Scene Builder.
- As you can see, it will be an empty file with only 'Border Pane' as the root to it. We can add as many controls, containers or widgets to it as we please and simultaneously, in the background, the FXML file will get automatically updated.

Now Scene Builder has been configured with Eclipse. Let's explore how to use it, and then create a simple interface to familiarise yourself with it. We will also integrate the application logic with the interface as we go along.

How to use Scene Builder

In the configuration section above, we have created a *Main.fxml* file in our JavaFX project 'MyFirstSceneBuilder'. Continuing from there, let's create a basic GUI application with the use of Scene Builder.

- Open your *Main.fxml* file with Scene Builder. You will see an empty centre section because you have not specified the minimum length and breadth of your border pane. Select the border pane from the hierarchy area, and set the minimum height and minimum width as 400 and 150 pixels, respectively, from the layout area.
- Now search for 'textfield' and 'button' from the library section, and add these to the 'CENTRE' and 'RIGHT' area of the border pane.
- Select *textfield* and set the margin as 5 units from the layout area in the *Inspector* section. Repeat the same process for the 'button'. Also, change the text of the button to 'Click Me'.